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Agentic AI and secure communications

Agentic AI, Autonomous Self-Healing, and NIST FIPS 203 (ML-KEM): Secure Communications, Hybrid TLS 1.3 PQC, and Real-Time Recovery

Quick Reference

Key Findings Table

Theme	Key Findings	Quantitative Metrics	Citations
Agentic AI & Self-Healing	Agentic AI with autonomous self-healing improves threat detection, reduces recovery time, and system downtime. Multi-agent accountability and quantum-resilient protocols are critical.	Threat detection accuracy: up to 96.8%; Recovery time reduction: 75%; Downtime reduction: 60%	1 2 3 4 5
Hybrid TLS 1.3 PQC (ML-KEM)	ML-KEM hardware acceleration (FPGA/ASIC/AVX-512) reduces key encapsulation overhead and energy consumption. Hybrid frameworks balance security and latency.	FPGA lifecycle latency: 17.78 µs; ASIC: 332 MHz, low power; AVX-512: up to 1.64× speedup	6 7 8
MARL for Anomaly & HNDL Detection	Multi-agent RL (MARL) with PPO, explainability, and robust communication improves real-time anomaly and "harvest-now-decrypt-later" (HNDL) detection.	Detection accuracy: up to 99.95%; Communication efficiency: +30%; Recovery rate: +4.3%	9 10 11 12
Autonomous DRV Loops	DRV loops with AI-driven detection and RL-based recovery achieve low-latency, zero-touch session recovery. Standardization of recovery metrics is lacking.	MTTR: 1.8 s (best case); Recovery success: 96%; Latency reduction: 25–40%	13 14 15 16

Direct Answer

- **Hybrid TLS 1.3 PQC key encapsulation overhead**: FPGA-based ML-KEM achieves lifecycle latency of 17.78 µs; ASIC implementations offer the lowest area-time product and power; AVX-512 acceleration provides up to 1.64× speedup over AVX2. Hybrid frameworks show negligible performance penalties compared to PQC-only protocols, supporting real-time secure communications in constrained environments 6 7 8 17.
- **Quantitative recovery latency**: DRV loop frameworks report MTTR as low as 1.8 seconds, with recovery latency reductions of 25–40% over baselines and recovery success rates up to 96% 13 14 15.

Study Scope

- **Time Period**: 2019–2024 (emphasis on post-2022 for PQC and 6G/AI developments)
- **Disciplines**: Cybersecurity, AI/ML, cryptography, hardware design, network management
- **Methods**: Systematic literature review, meta-analysis, benchmarking, simulation, hardware/software co-design, empirical evaluation

Assumptions & Limitations

- **Assumptions**: Hardware acceleration results are generalizable across similar platforms; MARL frameworks are evaluated under representative network and threat models; DRV loop metrics are comparable across domains.
- **Limitations**: Lack of standardized recovery latency metrics; limited real-world deployment data for 6G/zero-touch orchestration; some results are based on simulation or prototype hardware.

Suggested Further Research

- Standardization of recovery latency and resilience metrics for DRV loops across domains.
- Robust, interpretable, and energy-efficient multi-agent accountability frameworks for agentic AI with quantum-resilient protocols.
- Advanced intent conflict resolution and resource optimization in hierarchical closed-loop 6G network management.
- Stable convergence and adversarial robustness in MARL for real-time anomaly and HNDL detection.

1. Introduction

The convergence of agentic AI, autonomous self-healing, and post-quantum cryptography (PQC)—notably NIST FIPS 203 (ML-KEM)—is reshaping secure communications. As quantum threats and adversarial attacks evolve, the integration of hybrid TLS 1.3 PQC, multi-agent reinforcement learning (MARL) for anomaly detection, and zero-touch recovery mechanisms becomes essential for resilient, real-time systems [1] [6] [9] [13] . This review synthesizes recent advances and research gaps in these domains, focusing on practical and theoretical insights for next-generation secure communication infrastructures.

Scope and Motivation

This report covers:

- Agentic AI and self-healing for adaptive, accountable, and quantum-resilient security.
- ML-KEM and hybrid TLS 1.3 PQC key encapsulation, with a focus on hardware/software optimizations and overhead.
- MARL for real-time anomaly and "harvest-now-decrypt-later" (HNDL) detection.

- Autonomous Detection-Remediation-Verification (DRV) loops for zero-touch session recovery.
- Quantitative recovery latency metrics and open research challenges.

The motivation is clear: as quantum and adversarial threats outpace traditional defenses, advanced, autonomous, and explainable frameworks are needed to ensure secure, resilient communications [1] [6] [9].

2. Theoretical Frameworks

2.1 Agentic AI and Autonomous Self-Healing for Secure Communications

Advancements in Agentic AI and Self-Healing Systems

- **Agentic AI** integrates persistent memory, tool use, and multi-agent collaboration, enabling dynamic, autonomous decision-making [1] [18].
- **Self-healing mechanisms** (semantic checkpointing, cognitive rollback) reduce recovery latency and failure propagation in distributed systems [3].
- **Quantum-resilient defense**: Agentic AI frameworks increasingly incorporate quantum-resistant protocols and ethical governance to counter post-quantum threats [4] [5].
- **Empirical results**: Threat detection accuracy up to 96.8%, recovery time reduction by 75%, and downtime reduction by 60% over manual approaches [1].

Multi-Agent Security and Accountability

- **Explainable AI detection** and decentralized auditing (e.g., Hyperledger, blockchain) enhance transparency and traceability in multi-agent systems [19] [20].
- **Lifecycle-aware defense layers** (MAAIS) and modular, layered security frameworks maintain confidentiality, integrity, availability, and accountability [21].
- **Adaptive accountability**: Real-time tracing of responsibility flows and emergent norm detection prevent collusion and support ethical governance [20] [22].

Scalability, Robustness, and Energy Sustainability

- **Communication protocols** (AutoGen, LangChain, MCP, A2A, ANP) and virtualization frameworks support scalable, robust agent interactions [23] [24].
- **Energy-aware coordination**: Hybrid metaheuristics (Ant Colony, Firefly, PSO) and federated learning reduce energy consumption by up to 55.4% while maintaining operational fidelity [25] [26].
- **AI-blockchain integration**: Enhances both cybersecurity and energy sustainability in decentralized multi-agent systems [27].

Quantum-Resilient Defense Mechanisms

- **Hybrid agentic quantum-AI frameworks**: Combine quantum machine learning with agent-based decision-making, improving intrusion detection by 42% and reducing response latency by 55% [4].
- **QRANO and QR-MCP**: Lattice-based PQC and AI-driven anomaly detection reduce network attacks by 70% and improve resource utilization by 28.9% [5] [28].
- **Adaptive policy management**: Post-quantum multi-factor authentication and QKD/PQC hybridization ensure resilience and low latency [29] [30].

Synthesis: Agentic AI frameworks, when combined with self-healing and quantum-resilient mechanisms, offer adaptive, scalable, and robust security for next-generation networks. However, challenges remain in standardizing accountability, energy metrics, and integrating ethical governance at scale [31] [32].

2.2 NIST FIPS 203 (ML-KEM) and Hybrid TLS 1.3 PQC Key Encapsulation

ML-KEM Overhead and Performance in Hybrid TLS 1.3

- **ML-KEM** (Kyber) is the NIST-standardized KEM for PQC, offering compact key sizes and IND-CCA2 security [33].
- **Hardware acceleration** (FPGA/ASIC): FPGA lifecycle latency of 17.78 µs; ASICs achieve the best area-time product and lowest power, supporting all security levels [6] [8].
- **Hybrid TLS 1.3**: Integrating ML-KEM with classical KEMs shows negligible performance penalties, supporting real-time applications [17] [34].

Hardware Optimizations and Side-Channel Resistance

- **NEON/AVX-512 acceleration**: NEON on ARMv8-A reduces CPU cycles by up to 3.24×; AVX-512 achieves up to 1.64× speedup over AVX2 [7] [35].
- **FPGA/ASIC optimizations**: High-level synthesis, conflict-free memory mapping, and unified polynomial arithmetic modules reduce execution time and resource usage [36] [37].
- **Side-channel mitigation**: Masking, Isolated Wave Dynamic Differential Logic (IWDDL), and compiler-based randomization improve resistance to DPA/CPA attacks [38] [39].

Number Theoretic Transform and Rejection Sampling

- **NTT and rejection sampling**: Major contributors to computational overhead; optimized architectures (e.g., PQIns) and deterministic packetization reduce latency and improve resource efficiency [6] [40].
- **Rejection sampling**: Increases sampling time by 25–65% over CBD; architecture-aware optimizations can mitigate this [41] [42].

Hybrid TLS 1.3 PQC Frameworks and Adaptive Parameter Tuning

- **Dynamic adaptation**: Hybrid frameworks (e.g., PQTLS-AD, AQP-FDH) offload PQC certificate transmission, adapt cryptographic parameters, and optimize handshake latency [30] [43].

- **Dual algorithm certificates**: Enable rapid PQC deployment with backward compatibility for IoT/OT [44].
- **Empirical results**: Hybrid key exchange increases handshake latency and energy modestly compared to classical-only, but is the most practical near-term approach [45].

Comparative Latency and Energy Efficiency

- **FPGA/ASIC**: Offer lowest latency and energy consumption (e.g., 0.31 μJ per operation at 150 MHz for space-grade FPGA) [46].
- **AVX-512**: Provides flexibility and ease of deployment on CPUs, but higher power usage [47].
- **UAV/ICS applications**: ML-KEM encapsulation is a minor portion of total latency (3.9–8.0%); end-to-end encryption latency as low as 8.7 ms and energy overhead of 11.6 mJ per session [48] [49].

Synthesis: ML-KEM, when hardware-accelerated and integrated into hybrid TLS 1.3, enables practical, low-latency, and energy-efficient post-quantum secure communications. Optimizations in NTT, rejection sampling, and side-channel resistance are critical for deployment in constrained environments [6] [50] [51].

2.3 Multi-Agent Reinforcement Learning for Real-Time Anomaly and HNDL Detection

MARL Techniques for Anomaly and Attack Detection

- **MARL frameworks**: Detector and Classifier agents using Random Forest Q-value estimation achieve up to 99.95% accuracy in real-time intrusion detection [9].
- **Hybrid RL**: DDQN and MADDPG, enhanced by feature engineering and PPO, outperform existing RL-based models in detection accuracy and computational efficiency [52].
- **Stacked-LSTM ensembles**: Improve detection of adversarial attacks in multi-agent systems [53].

Handling Partial Observability and Communication Constraints

- **EMAC, GRDC, P-MARL**: Use graph networks, local interaction graphs, and probabilistic reasoning to improve efficiency and robustness under partial observability and communication constraints [10] [54] [55].
- **Semantic communication**: Enhances information sharing and coordination in unreliable environments [56].

Feature Engineering, PPO, and Explainability

- **PPO integration**: Improves exploration, coordination, and stability; combined with feature engineering, achieves up to 99.19% accuracy and 97.97% F1 score [11] [57].
- **Explainable AI (XAI)**: SHAP, LIME, GradCAM, and rule induction methods enhance interpretability and trust in MARL-based detection systems [11] [58].

Stable Convergence and Robustness Against Adversarial Attacks

- **CTDE, robust Q-learning**: Address convergence and stability, but state-adversarial attacks remain a challenge [59] [60].

- **Adversarial training (WALL, ROMAT)**: Improves robustness against coordinated and stealthy attacks, with up to 4.3% performance gain over traditional methods [12] [61].

Communication-Efficient MARL and Safe Reinforcement Learning

- **Event-triggered and information bottleneck strategies**: Reduce bandwidth usage and maintain performance under constraints [62] [63].
- **Safe RL**: Control Barrier Functions and dynamic shielding provide formal safety guarantees and resilience [64] [65].

Synthesis: MARL frameworks, when enhanced with PPO, explainability, and robust communication, offer high-accuracy, real-time anomaly and HNDL detection. However, stable convergence and adversarial robustness under partial observability remain open challenges [66] [67].

2.4 Autonomous Detection-Remediation-Verification (DRV) Loops and Zero-Touch Session Recovery

DRV Loop Frameworks and Recovery Latency Metrics

- **AI-driven DRV loops**: Achieve 97% anomaly detection, 96% recovery success, and MTTR of 1.8 seconds [13].
- **Goal-oriented Communication (GoC)**: Reduces fault detection and recovery time by up to 82.6% [68].
- **Zero-Touch GenAI Coach**: Reduces MTTR by 37% in CI/CD pipelines [69].

Standardized Metrics and Benchmarking

- **Metrics**: MTTR, resiliency score, detection latency, diagnosis time, reconfiguration success, and recovery rate [15] [70].
- **Benchmarking**: Edge-IIoTset dataset and explainable GNNs provide standardized performance metrics (e.g., 96.3% fault detection, 25–40% latency reduction) [14].

Real-Time Adaptability and Drift Management in DRL-Based Recovery

- **Composite Drift Index**: Proactively detects DRL performance degradation, improving detection accuracy by 23.4% [71].
- **Hybrid neural networks (LSTM)**: Capture time dependencies for proactive failure recovery [72].
- **Continual learning**: Intentional forgetting and dynamic batch sizing improve adaptation and reduce healing time [73] [74].

Hierarchical Closed-Loop Automation and Intent Management in 6G

- **LLM-based intent management**: Detects and resolves conflicts, improving intent fulfillment to ~83.89% [75] [76].
- **Multi-agent architectures**: Enhance orchestration speed by 20% and optimize resource allocation [77].

- **Game-theoretic and ML-based conflict resolution**: Prevents performance degradation and improves reliability [78] [79].

Autonomous Vulnerability Remediation in Zero-Trust Architectures

- **AI-driven remediation**: Reduces MTTR by 81% (to ~2.7 days), with high detection and recovery rates [13] [80].
- **Best practices**: Continuous verification, least-privilege access, automated patch orchestration, and dynamic trust scoring [81] [82].

Balancing Detection Accuracy, Remediation Success, and Recovery Latency

- **RL-based recovery**: Safe RL and recovery-based shielding maintain safety and accelerate convergence [83] [84].
- **Accelerated adaptation**: Diagnostic error signals enable up to 80× faster recovery in nonlinear systems [85].
- **Multi-agent RL**: Dynamically adapts strategies to reduce latency and improve reliability [86].

Synthesis: Autonomous DRV loops, when integrated with AI-driven detection, RL-based recovery, and hierarchical intent management, achieve low-latency, zero-touch session recovery. However, standardization of recovery metrics and real-time adaptability in dynamic environments require further research [71] [75].

3. Methods & Data Transparency

- **Systematic Literature Review**: Aggregated findings from peer-reviewed articles, standards, and technical reports (2019–2024).
- **Meta-Analysis**: Quantitative synthesis of performance, latency, and accuracy metrics across domains.
- **Benchmarking**: Use of standardized datasets (e.g., Edge-IIoTset), hardware prototypes (FPGA/ASIC/AVX-512), and simulation environments for empirical validation.
- **Transparency**: All claims are supported by explicit citations; limitations and assumptions are clearly stated.

4. Critical Analysis of Findings

4.1 Agentic AI and Self-Healing

- **Strengths**: High adaptability, explainability, and quantum-resilience; significant reductions in recovery time and downtime.
- **Weaknesses**: Multi-agent accountability and energy sustainability frameworks are still maturing; standardization is lacking.
- **Opportunities**: Integration with quantum communication and ethical governance frameworks.

4.2 ML-KEM and Hybrid TLS 1.3 PQC

- **Strengths**: Hardware acceleration (FPGA/ASIC/AVX-512) enables real-time, energy-efficient secure communications; hybrid frameworks balance security and latency.
- **Weaknesses**: NTT and rejection sampling remain computational bottlenecks; side-channel resistance requires ongoing innovation.
- **Opportunities**: Further optimization for IoT/edge devices and integration with adaptive, intent-driven network management.

4.3 MARL for Anomaly and HNDL Detection

- **Strengths**: High detection accuracy, robust to partial observability and communication constraints; explainability enhances trust.
- **Weaknesses**: Stable convergence and adversarial robustness are not fully solved; real-world deployment is limited.
- **Opportunities**: Unified frameworks combining MARL, XAI, and hardware-accelerated cryptography.

4.4 Autonomous DRV Loops

- **Strengths**: Low-latency, high-success recovery; hierarchical intent management improves reliability.
- **Weaknesses**: Lack of standardized recovery metrics; intent conflicts and resource bottlenecks in 6G remain challenging.
- **Opportunities**: Standardization, advanced conflict resolution, and integration with zero-trust architectures.

5. Real-World Implications

- **Critical Infrastructure**: ML-KEM-accelerated hybrid TLS 1.3 and agentic AI frameworks are deployable in UAV, ICS, and IoT environments, supporting real-time, quantum-resilient secure communications [49] [87].
- **Zero-Touch Networks**: Autonomous DRV loops and hierarchical intent management are foundational for 6G zero-touch orchestration, reducing operational overhead and improving resilience [75] [88].
- **Cyber Defense**: MARL-based anomaly and HNDL detection systems, when explainable and robust, can be integrated into SOCs for proactive, real-time threat mitigation [9] [11].
- **Energy Sustainability**: Energy-aware coordination and hardware optimizations ensure that security enhancements do not compromise sustainability in large-scale, distributed systems [25] [89].

6. Future Research Directions

- **Standardization of Recovery Metrics**: Develop and adopt standardized, domain-agnostic metrics for recovery latency, resilience, and DRV loop performance [70] [75].

- **Stable, Interpretable MARL**: Advance MARL algorithms for stable convergence, adversarial robustness, and explainability in real-time, partially observable environments [66] [67].
- **Scalable Multi-Agent Accountability**: Design robust, low-energy, and interpretable accountability frameworks for agentic AI, integrating quantum-resilient protocols and ethical governance [31] [32].
- **Hierarchical Intent Management**: Innovate in intent conflict resolution, resource optimization, and explainable orchestration for 6G zero-touch networks [75] [88].
- **Unified Adaptive Security Frameworks**: Explore coupling MARL, XAI, and hardware-accelerated PQC for adaptive, autonomous, and quantum-resilient cybersecurity systems [4] [11] [30].
- **Real-World Deployment and Benchmarking**: Expand empirical studies and benchmarking in operational environments, especially for DRV loops and hybrid PQC frameworks in critical infrastructure [14] [49].

Synthesis: The literature converges on the need for adaptive, agentic AI frameworks that integrate robust self-healing, quantum-resilient cryptography, and explainable, multi-agent learning for secure, real-time communications. Hardware acceleration and hierarchical automation are enabling practical deployments, but standardization, stability, and interpretability remain key research frontiers [1] [6] [9] [13].

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